Bruno Garcia

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Education

University of Michigan Ann Arbor, MI

Bachelor of Science, Computer Science — GPA: 3.62 *May 2028*

- **Coursework:** Data Structures and Algorithms, Object Oriented Programming, Discrete Math, Web Systems, Computer Organization, Software Engineering, Foundations of Computer Science
- **Extracurriculars:** Michigan Climbing Team (National Qualifier), Michigan Table Tennis, MHacks

Experience

Cambridge Mobile Telematics - *Software Engineer Intern* June 2026 – August 2026

- Cached repeated table initialization every 6 hours for the telematics analytics API, reducing database calls by **∼35%** and speeding up dashboard and chart queries across large customer datasets.
- Built and deployed MCP tools with Terraform that let an AWS DevOps agent query a time-series metrics database for anomalies, reducing root-cause analysis time by **∼80%** for ride failures and device-update issues.
- Engineered an event-driven Python service using webhooks to refresh analytics charts when source data changed, serving **1,000+** daily chart requests while eliminating redundant backend calls and reducing compute costs.
- Implemented a simple rule based anomaly-detection algorithm in Python using pandas and DuckDB to scan **400,000+** driver records across rolling 3-day windows and surface ride-failure patterns for further investigation.
- Authored **20+ evaluations** to test the AWS DevOps agent's response accuracy, tool selection, and ability to complete ride-failure investigation tasks.

Michigan Medicine - *Software Engineer Intern* May 2025 – Aug 2025

- Built an internal pharmaceutical analytics platform with Python, FastAPI, and PostgreSQL, allowing researchers to explore dispensing trends, and filter results without writing SQL or manually exporting data for analysis.
- Reduced dashboard load time by **60%** across **500K+ dispensing records** using PostgreSQL indexes, and Redis caching, cutting repeated database work and making medication and cohort views faster to use.
- Improved platform reliability with FastAPI validation, PostgreSQL and access controls, catching bad dispensing records, restricting study access, and reducing errors researchers encountered using the internal analytics platform.
- Automated the ingestion and cleaning of pharmaceutical records, standardizing inconsistent fields and reducing manual data preparation time by **200%** before analysis.

Projects

Dora (MHacks 2026) | *Python, React, TypeScript, PostGIS, Rasterio, GLiClass, Docker*

- Engineered a centralized dashboard for researchers studying animal migration and population patterns using an **agentic pipeline** coordinating 4 specialized roles for request interpretation, data retrieval, dataset preparation, and reporting, connecting natural-language queries to research papers, animal GPS records, and satellite imagery.
- Integrated PostGIS spatial queries and Rasterio image processing into agent workflows and built an interactive React, TypeScript, and Leaflet mapping interface, saving researchers **100+** hours of manual dataset preparation and exploration.

PomodoroLB | *JavaScript, Node.js, Next.js, PostgreSQL, Supabase, Vercel*

- Deployed a Next.js Pomodoro app where users could track study sessions and compare study time through weekly, monthly, and all-time leaderboards, averaging **20+ active users** and 50+ hours logged.
- Added user accounts and session tracking with Supabase and PostgreSQL, ensuring completed study sessions were recorded correctly and reflected in leaderboard rankings.

Fortnite Item Tracker | *Golang, Gin, PostgreSQL, Docker, Discord API*

- Built and deployed a Discord bot and web app with alerts for tracked cosmetics in the Fortnite shop.
- Implemented Discord OAuth2 authentication and JSON Web Tokens for secure session management.

Skills

Languages: Python, C++, Go, TypeScript, JavaScript, SQL, C#, R

Tools: Git, Node.js, SQLite, PostgreSQL, Redis, Docker, Jenkins, Terraform, REST APIs

Frameworks/Libraries: React, Django, Flask, Apache Spark, pandas